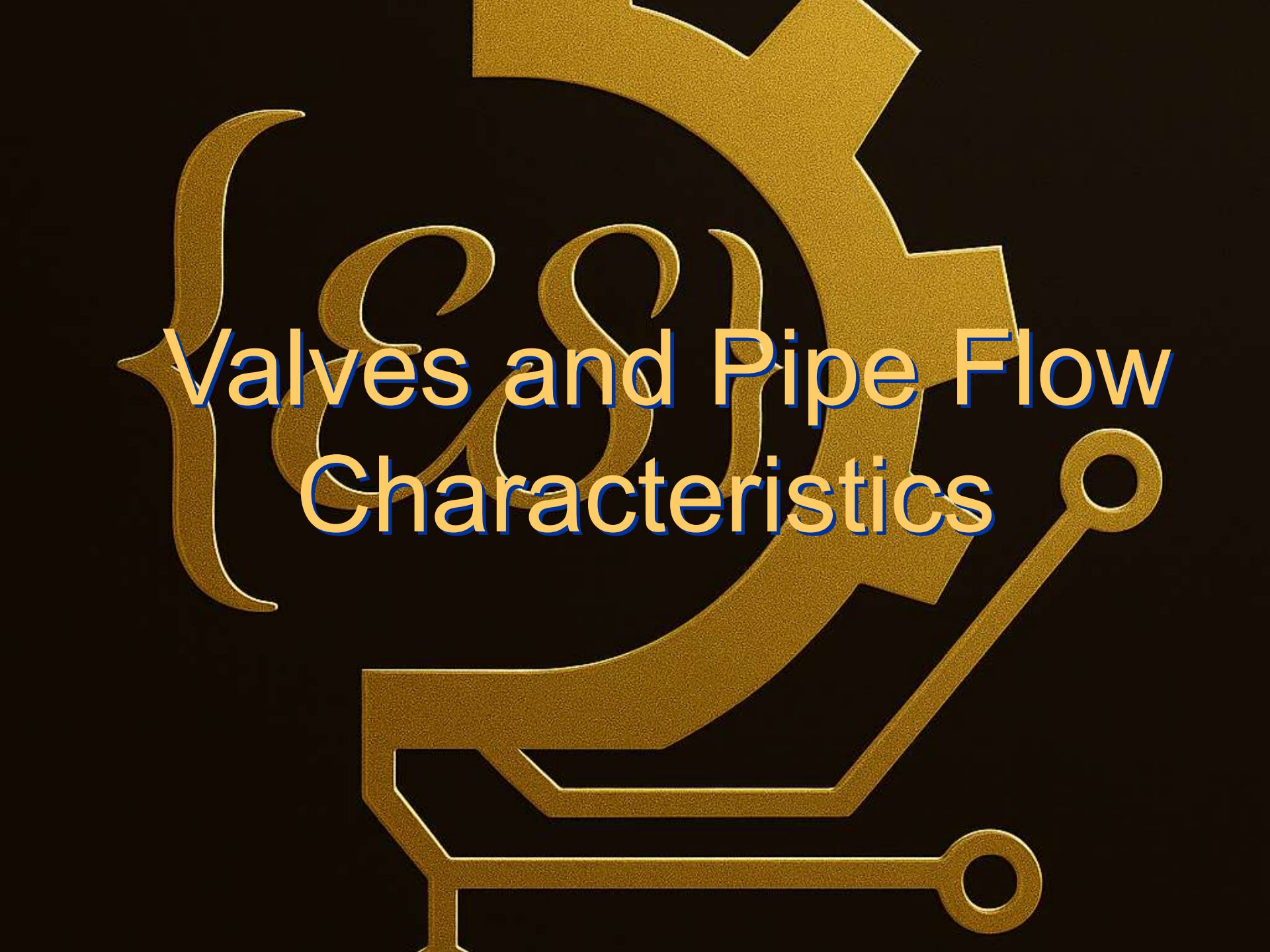
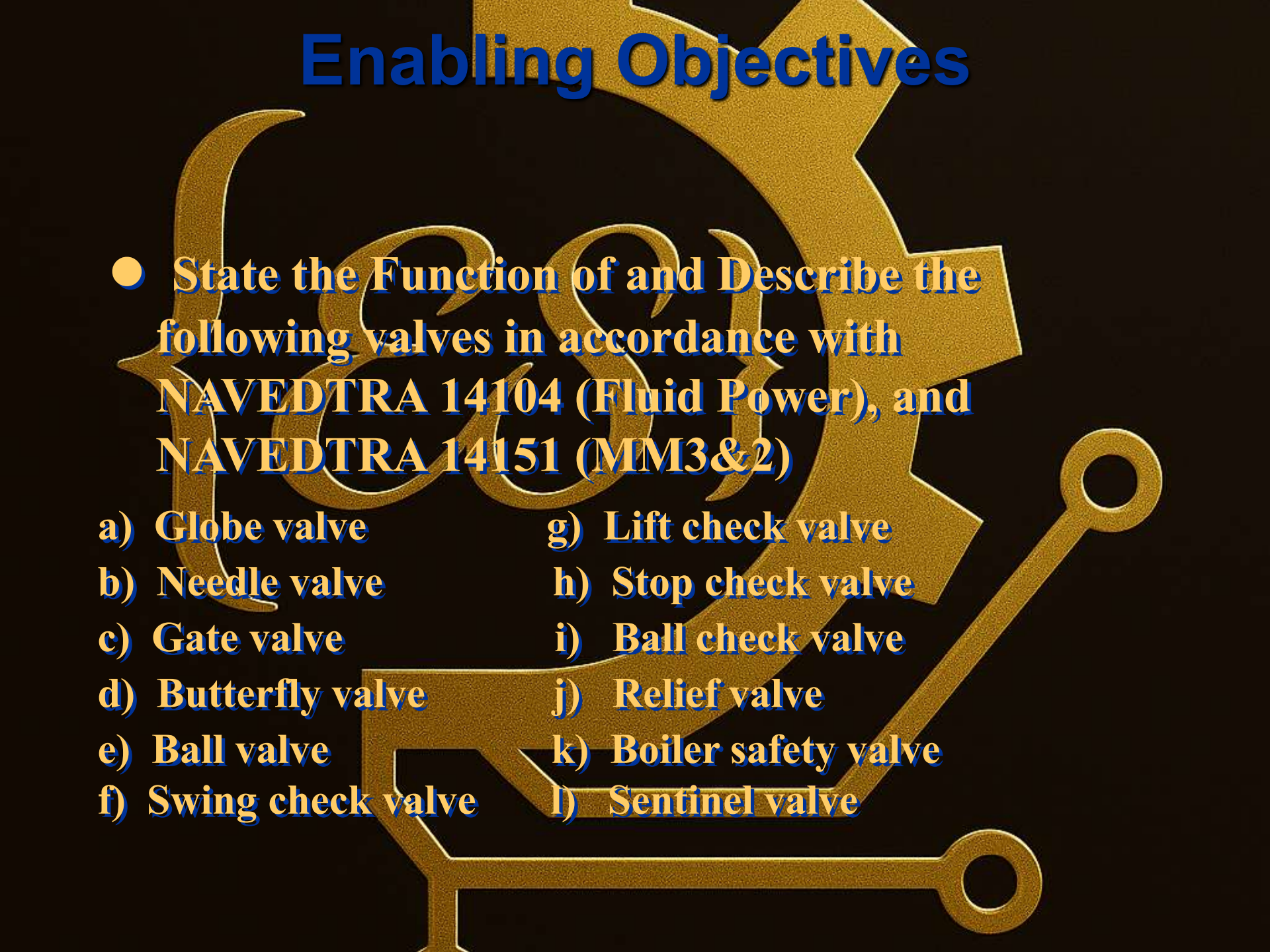


The background features a large, intricate, golden-colored mechanical design. It includes a large gear at the top, a complex valve or piston mechanism in the center, and a series of interconnected pipes and fittings at the bottom. The entire design is rendered in a metallic, brushed-gold texture against a solid black background.

# Valves and Pipe Flow Characteristics

# Enabling Objectives



- State the Function of and Describe the following valves in accordance with NAVEDTRA 14104 (Fluid Power), and NAVEDTRA 14151 (MM3&2)

- |                      |                        |
|----------------------|------------------------|
| a) Globe valve       | g) Lift check valve    |
| b) Needle valve      | h) Stop check valve    |
| c) Gate valve        | i) Ball check valve    |
| d) Butterfly valve   | j) Relief valve        |
| e) Ball valve        | k) Boiler safety valve |
| f) Swing check valve | l) Sentinel valve      |

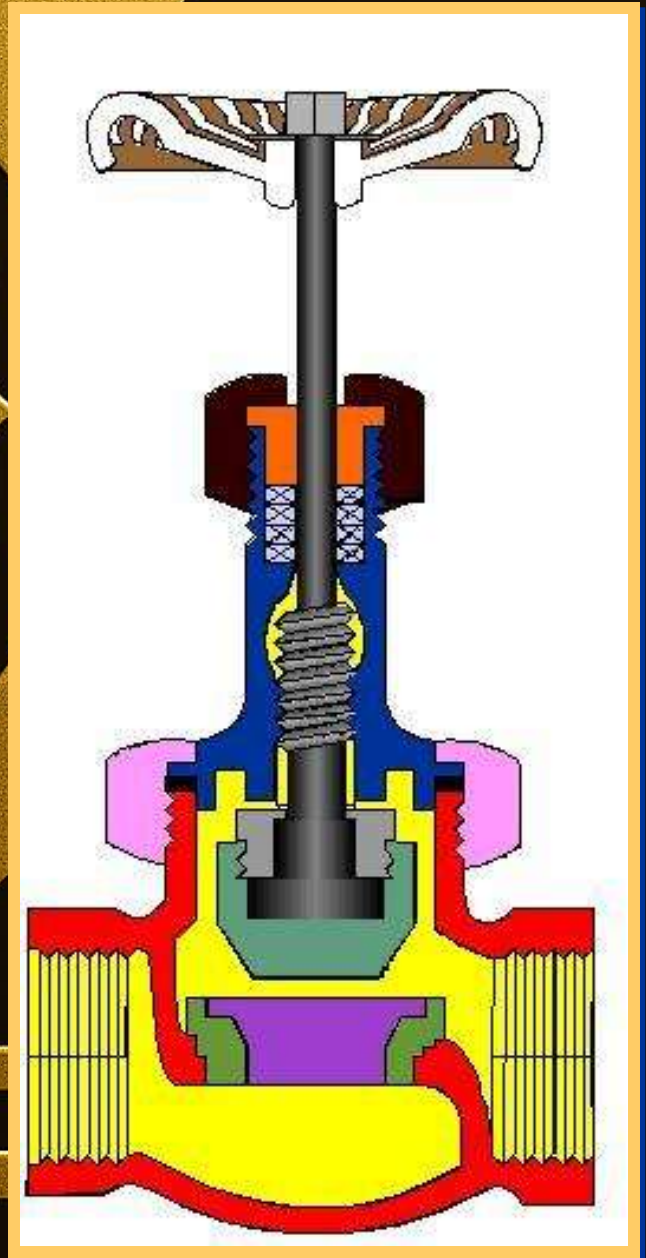


# Enabling Objectives

- Understand how a pipe-flow apparatus establishes a number of different flow paths and how certain fittings/components affect pressure throughout a piping system.

# GLOBE VALVE

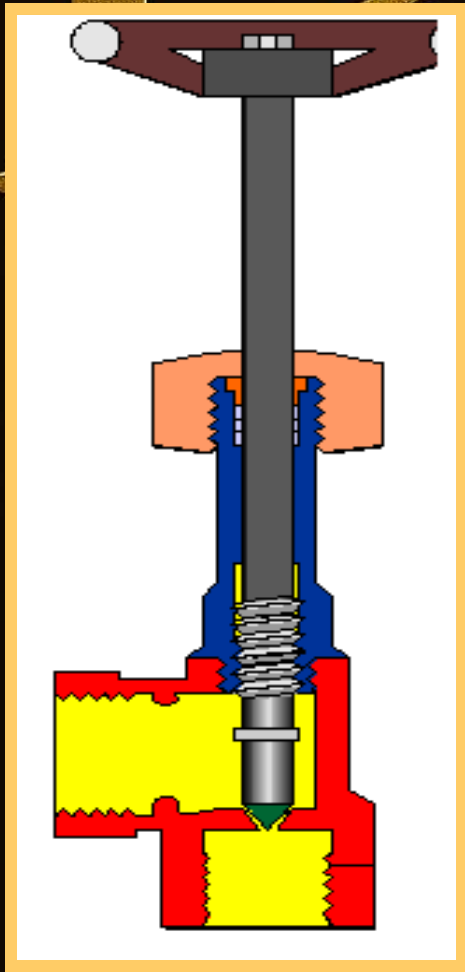
- One of the most common types found onboard a naval vessel
- Used for isolation and throttling





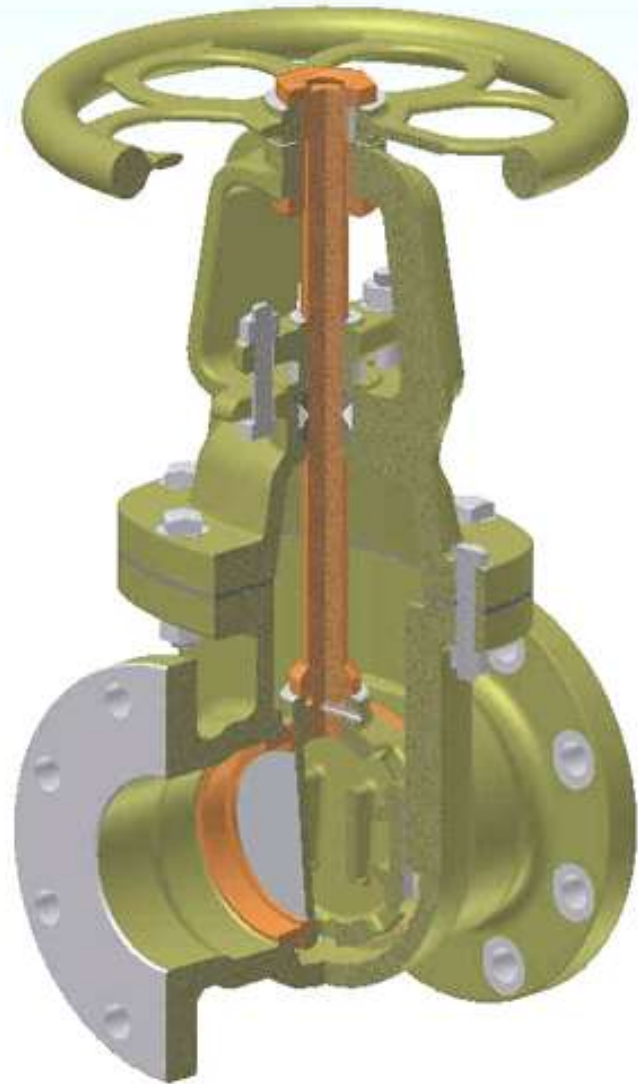
# NEEDLE VALVE

- Similar to the Globe Valve
- Disk is shaped like a needle
- Used where precise adjustment of flow is necessary



# GATE VALVE

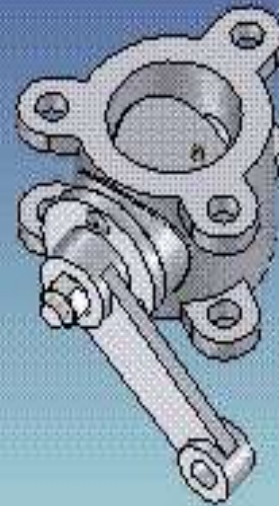
- Used when minimum flow restriction is needed
- Only used to isolate fluid flow, NEVER to throttle



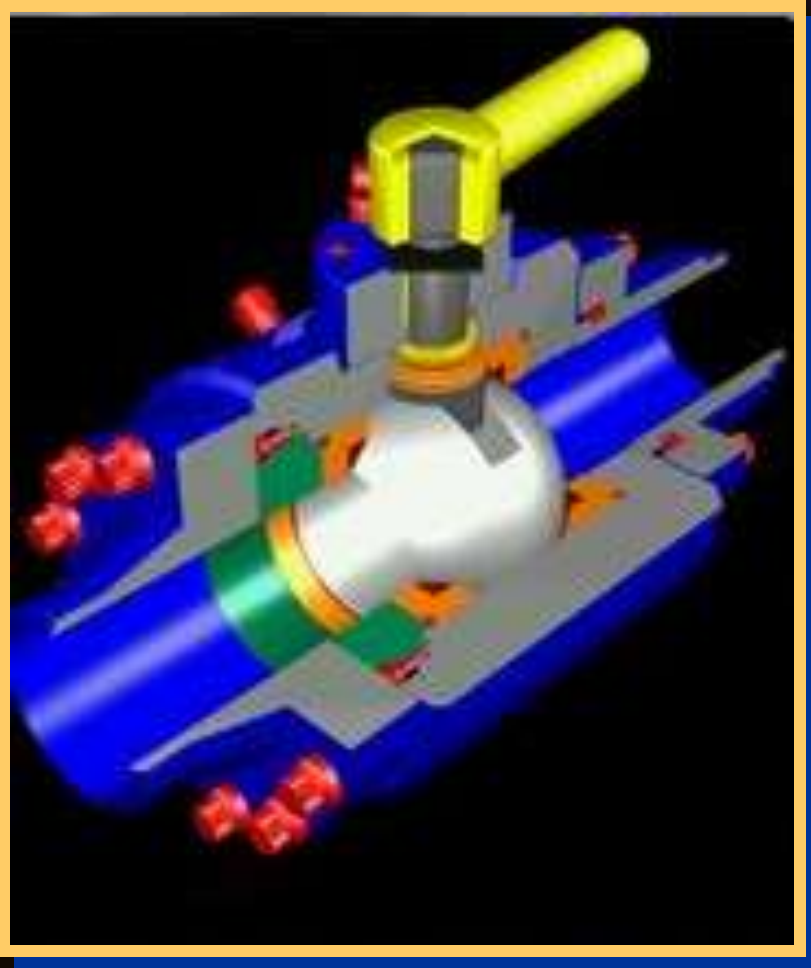


# BUTTERFLY VALVE

- Provides positive shut off
- Can be used for throttling, but not recommended
- To operate, turn handle  $\frac{1}{4}$  turn to rotate disk  $90^\circ$



# BALL VALVE

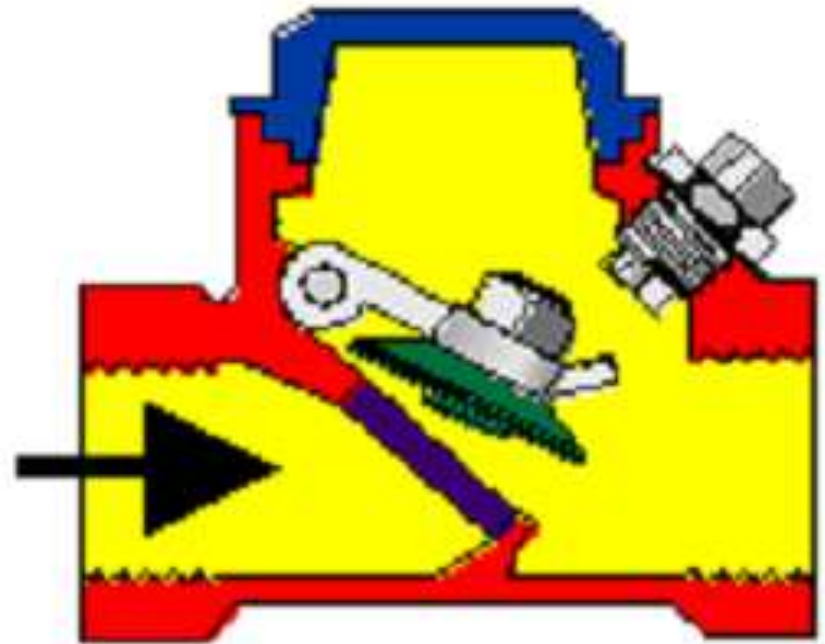


- Used for isolation
- Cannot be used for throttling
- Valve is operated by turning handle  $\frac{1}{4}$  turn to rotate ball  $90^\circ$
- Used in systems where minimal flow restriction is required

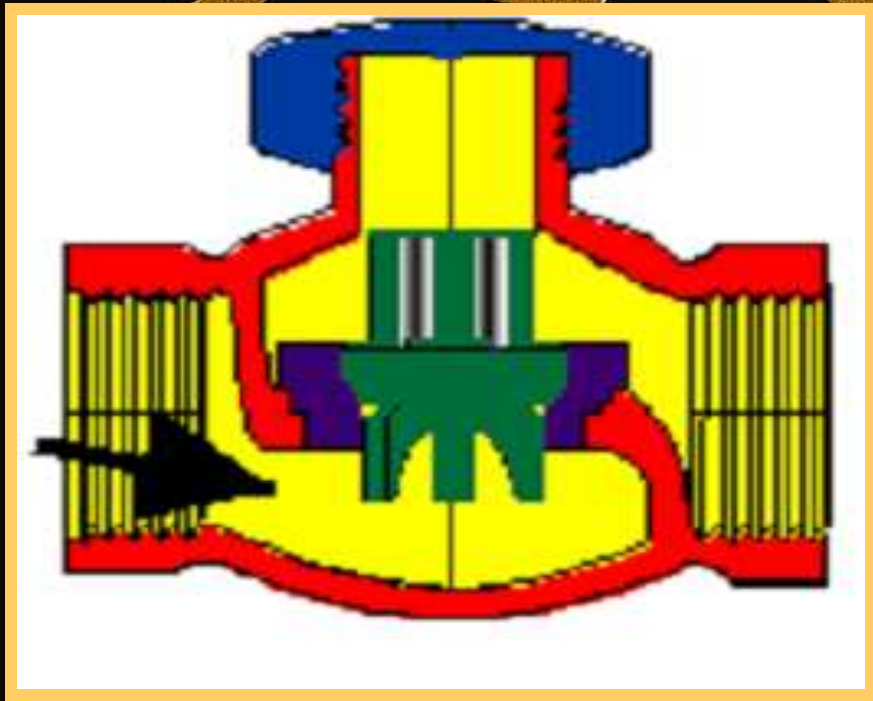


# SWING CHECK VALVE

- Fluid flow opens the hinged disk
- Cannot be mounted upside down



# LIFT CHECK VALVE

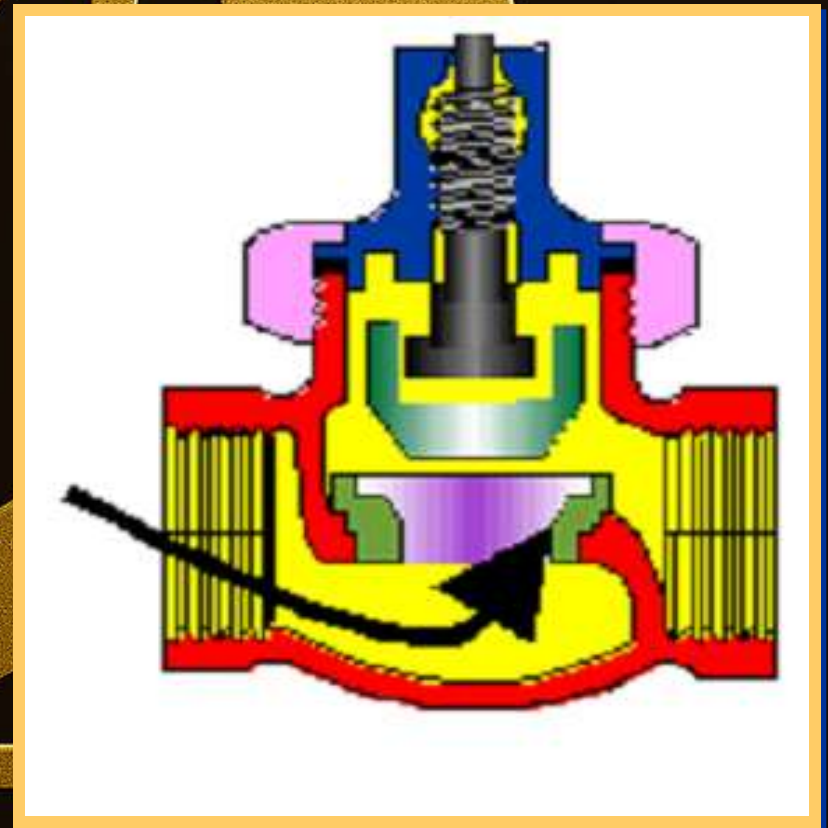


- Fluid flow lifts disk off of the seat
- Cannot be mounted upside down



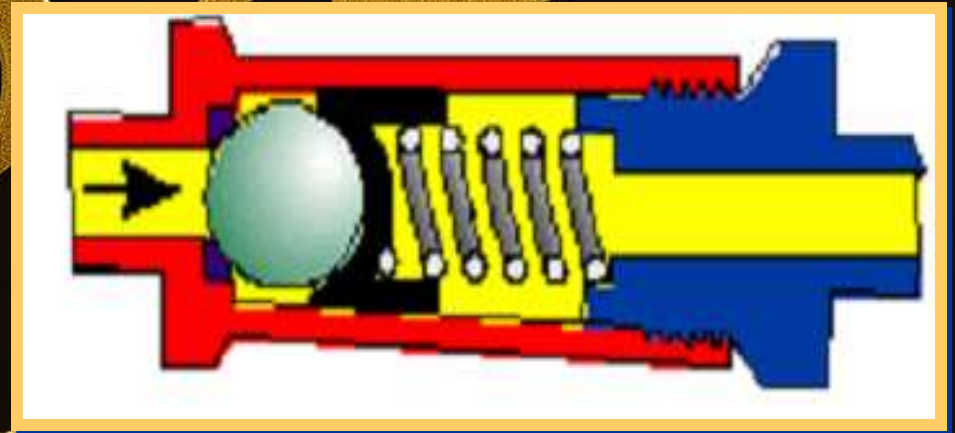
# STOP CHECK VALVE

- Acts as an isolation valve and check
- Similar to the Globe Valve, but the disk is not attached to the stem



# BALL CHECK VALVE

- UTILIZES A BALL WITH A SPRING INSTEAD OF A DISK
- CAN BE MOUNTED UPSIDE DOWN





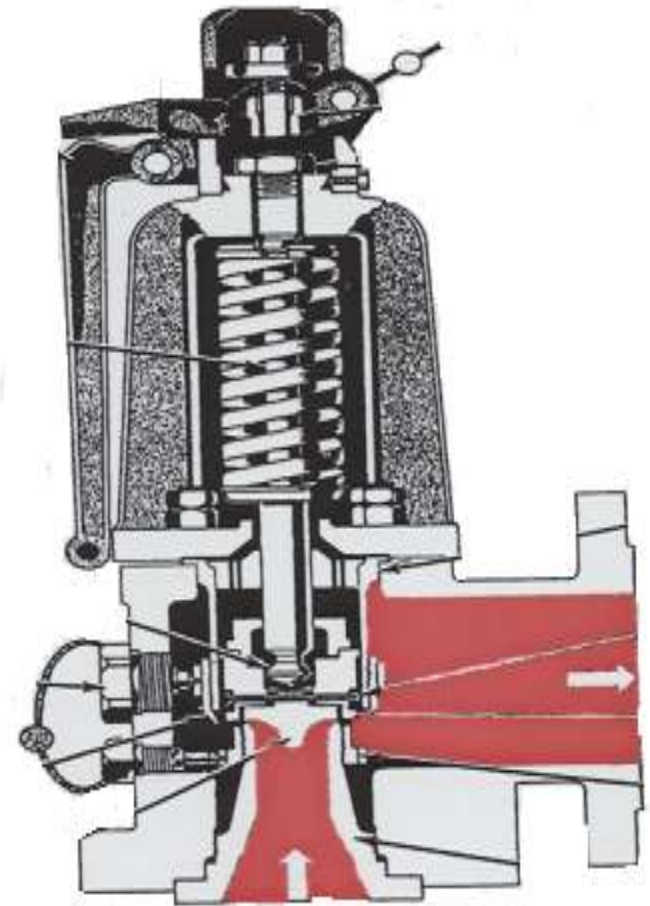
# RELIEF VALVE



- PROTECTS FROM EXCESSIVE PRESSURE
- OPENS AUTOMATICALLY AT A PRESET PRESSURE AND CLOSES AT A SLIGHTLY LOWER PRESSURE

# SAFETY VALVE

- Installed on boilers
- Reduces steam drum pressure to a safe operating point





# SENTINEL VALVE

- Warns of impending overpressure
- Does not relieve pressure

